

SOUNAK SARKAR

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[LinkedIn](#) | [GitHub](#) | [LeetCode](#)

SUMMARY

Final-year B.Tech CSE (AI & ML) student specializing in machine learning pipelines and agentic AI systems. Proficient in XGBoost, ensemble methods, and LLM frameworks (LangChain, LangGraph). Delivered production solutions including a 98% accurate cancer detection model and a multi-agent AI system. Qualified for ET GenAI Hackathon 2026 Phase 2 — among 6,000 teams selected out of 55,000+ applicants.

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **Core CS Concepts:** Object-Oriented Programming (OOP), Data Structures & Algorithms, Database Management Systems (DBMS)
- **Machine Learning:** Scikit-Learn, XGBoost, TensorFlow, Ensemble Methods, Model Evaluation
- **AI & LLM:** LangChain, LangGraph, LangSmith, OpenAI API, Groq API, Google Gemini, Prompt Engineering, RAG (Retrieval-Augmented Generation), Multi-Agent Systems, LLM Orchestration, HuggingFace
- **Data Science:** NumPy, Pandas, Matplotlib, Feature Engineering, Data Preprocessing, Statistical Analysis
- **Backend & APIs:** FastAPI, REST API, Pydantic, Docker
- **Tools & Deployment:** Git, GitHub, VSCode, Jupyter Notebook, Streamlit, Render, Vercel, API Integration, OCR (Tesseract), Tavily
- **Databases:** MySQL, Qdrant (Vector DB)
- **Soft Skills:** Team Collaboration, Analytical Thinking, Problem Solving, Technical Adaptability, Leadership

PROJECTS

Agent Mind — Advanced Multi-Model AI Agent | *Python, LangGraph, LangSmith, Streamlit* [GitHub](#)

- Architected a production-grade multi-tool AI agent system leveraging LangGraph StateGraph for intelligent query routing and orchestration across web search (Tavily), image generation (Pollinations), and document analysis.
- Developed an innovative dual-model comparison framework utilizing Mistral AI as a judge to evaluate and select optimal responses from Google Gemini and Groq API (Llama 3.1), improving response quality and reliability.
- Integrated advanced file analysis module with Tesseract OCR fallback supporting multiple file formats (.pdf, .py, .txt) for comprehensive document processing across varied use cases.
- Integrated Qdrant Cloud vector database with HuggingFace MiniLM-L6-v2 embeddings for RAG-based persistent long-term memory, enabling semantic retrieval of past interactions across agent sessions.
- Implemented LangSmith for distributed tracing and debugging, along with gTTS for text-to-speech conversion, delivering a seamless multi-modal user experience.
- Deployed production-ready interactive web application via Streamlit with multi-session support and real-time agent reasoning visibility.

Breast Cancer Detection System | *Python, XGBoost, Scikit-Learn, SHAP, Plotly, Streamlit* [GitHub](#)

- Engineered an end-to-end ML diagnostic system using XGBoost achieving 98% accuracy on the Wisconsin Breast Cancer dataset, with a full preprocessing pipeline (feature scaling, normalization, EDA) to ensure clinical-grade reliability.
- Integrated Explainable AI (SHAP) with waterfall charts to provide feature-level clinical decision breakdowns for every prediction, eliminating the black-box problem for medical stakeholders.
- Built interactive 3D scatter plots using Plotly to visualize clustering of benign vs. malignant cases across 30 high-dimensional features, with a live ROC-AUC and Confusion Matrix evaluation dashboard.
- Deployed a production-ready Streamlit web app with a "Load Random Patient" feature for instant real-world testing, attracting 64 clones from 41 unique users on GitHub.

EDUCATION

Dr. Sudhir Chandra Sur Institute of Technology & Sports Complex

Kolkata, India

Bachelor of Technology — Computer Science and Engineering (AI & ML) (CGPA: 7.83 / 10)	2022 – Present
Future Campus School	Kolkata, India
CBSE Class XII, Science Stream (80.4%)	2020 – 2022
Future Campus School	Kolkata, India
CBSE Class X (78%)	2018 – 2020

CERTIFICATIONS

- **AI Foundations Associate** | Oracle Cloud Infrastructure — Scored 78% in the certification exam Aug 2024
- **GenAI Powered Data Analytics Job Simulation** | Tata / Forage — Completed end-to-end data analytics simulation leveraging generative AI tools Aug 2024

ACHIEVEMENTS & ACTIVITIES

- **CS Mastermind Leaderboard — Rank 14** | The Vidyawan | Secured 14th rank overall competing alongside 78 talented minds from the computer science community.
- **ET GenAI Hackathon 2026 — Phase 2 Qualifier** | Ranked among top 6,000 teams from 55,000+ participants nationwide; built a multi-agent SCM system using LangGraph, Gemini 2.5 Flash, Groq, and Qdrant.
- **Open-Source Contributor** | Production-ready ML and AI application deployments publicly available on GitHub ([sounakss7](#)), including agentic AI systems and clinical ML models.
- **Academic Project Lead** | Led development of AGENT_MIND under the Neuroplexa AI brand — a multi-model agentic system with a four-layer security pipeline, mentored by Prof. Sanjoy Bhattacharya.
- **Technical Self-Learner** | Completed structured curricula in FastAPI, SQL, Docker, and LLM orchestration frameworks alongside full-time academic coursework.